AIRIN AKTER
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EDUCATION

Doctor of Philosophy (PhD), Geography 2023-present
Department of Geography, University of Florida, Gainesville, FL, USA
Advisor: Dr. Corene Matyas
Dissertation: Spatial Analysis of Precipitation Patterns Associated with Tropical Cyclones (ongoing)
Atmospheric Science Certificate (2025)

Master of Science (MS), Geography and Environment 2019
Department of Geography and Environment, University of Dhaka, Bangladesh
Socio-economic Impact of Embankment Breaching on the Community of Southkhali in Sharankhola, Bagerhat, Bangladesh

Bachelor of Science (BSc), Geography and Environment 2017
Department of Geography and Environment, University of Dhaka, Bangladesh

PROFESSIONAL EXPERIENCE

Graduate Assistant-Teaching August 2023 – Present
Department of Geography, University of Florida
GEO 2242 Extreme Weather (Instructor)
GEO 2230 Living with Rising Seas (Grader)

Regional Ambassador, #EverywhereSheMaps Program October 2020 – February 2023
YouthMappers (https://www.youthmappers.org/regional-ambassadors)
Built technical skills of young women in the YouthMappers network and helped them create inclusive research and mapping communities.
Provided training on ArcGIS, QGIS, InaSAFE, and OpenStreetMap.

Senior Research Associate August 2020 – December 2021
Capacity Building Service Group (CBSG), Bangladesh
Managed GIS data to analyze women's dietary systems in disaster-vulnerable areas of Bangladesh.

Planning Officer November 2019 – March 2020
Planning and Management Consultants (PMC), Bangladesh
Developed a Land Management System (LMS) for the Bangabandhu Bridge using GIS.

RESEARCH EXPERIENCE

PhD Dissertation: Spatial Analysis of Precipitation Patterns Associated with Tropical Cyclones 2023– Present
Advisor: Dr. Corene Matyas, Department of Geography, University of Florida
Paper 1: Compared IMERG and MSWEP satellite rainfall datasets for landfalling tropical cyclones (In preparation of journal submission)
Paper 2: Exploring the Influence of Environmental Conditions on Tropical Cyclone Rainfall Structure at Different Landfalling Stages
Paper 3: Atmospheric Moisture, Vertical Wind Shear, and Storm Motion to Tropical Cyclone Rainfall Structure: A Self-Organizing Map Analysis

Research Fellow, YouthMappers Research Fellowship March 2018 – April 2019

Project: Evacuation Plan during Cyclone Warning to Reduce Vulnerability of Women in Rayenda, Sharankhola, Bagerhat, Bangladesh

Built a native-language mapping and evacuation plan for community resilience.

Mentor: Dr. Nahid Rezwana, Associate Professor, University of Dhaka.

PUBLICATIONS

Peer-reviewed book chapter

Akter, A., & Tasnim, M. (2023). Mapping for Women's Evacuation Plans During Climate-Induced Disasters. In P. Solís & M. Zeballos (Eds.), *Open Mapping towards Sustainable Development Goals*. Sustainable Development Goals Series. Springer, Cham. https://doi.org/10.1007/978-3-031-05182-1_22

HONORS AND AWARDS

2026 University of Florida (UF) CLAS Graduate Studen t Travel Fund, American Geophysical Union (AGU) 2026 Annual Meeting, San Francisco, CA (December 7–11; upcoming)

2026 NSF NCAR MMM CELCIUS Graduate Visitor Program, Boulder, CO (July 26–31). Selected to present research and meet NCAR scientists.

2026 American Meteorological Society (AMS) Student Travel Fund, 106th AMS Annual Meeting, Houston, TX ((January 25–29)

2025 John & Fawn Dunkle Award for Graduate Student Travel (American Association of Geographers (AAG) 2025 and American Geophysical Union (AGU) 2025)

2025 University of Florida (UF) CLAS Graduate Student Travel Fund; UF Graduate Student Council Award; UF Department of Geography Conference Travel Award (American Geophysical Union (AGU) 2025)

2024 University of Florida (UF) Water Institute Travel Award; UF Graduate Student Council Award; UF Department of Geography Conference Travel Award (American Association of Geographers (AAG) 2024)

2022 YouthMappers Conference Scholarship, State of the Map and Free and Open-Source Software for Geospatial (FOSS4G), Florence, Italy

2018–2019 YouthMappers Research Fellowship, funded by YouthMappers and USAID through the GeoCenter (Award #AID-OAA-G-15-00007); hosted at The George Washington University and West Virginia University (June 2018)

2018 YouthMappers conference Scholarship, 4th State of the Map Asia, Bengaluru, India

2017 YouthMappers conference Scholarship, State of the Map Asia, Kathmandu, Nepal

PRESENTATIONS

Dec 2026 Influence of Environmental Conditions on Tropical Cyclone Rainfall Structure at Different Landfalling Stages. **AGU Annual Meeting, San Francisco, CA**. (Poster) (upcoming).

Jul 2026 Spatial Analysis of Tropical Cyclone Precipitation Structure: From Datasets to Drivers. **NSF NCAR, UCAR, and UCP Summer Student Conference, Boulder, CO**. (poster)

Dec 2025 Comparative Analysis of IMERG and MSWEP Dataset in Capturing Tropical Cyclone Rainfall. **AGU Annual Meeting, New Orleans, LA.** (Poster)

Mar 2025 Comparative Analysis of IMERG and MSWEP Dataset in Capturing Tropical Cyclone Rainfall for 9 Cases. **AAG Annual Meeting, Detroit, MI**. (Paper presentation)

Nov 2024 Comparative Analysis of IMERG and MSWEP Dataset in Capturing Tropical Cyclone Rainfall **Southeast Division of the AAG (SEDAAG), Greenville, SC** (Paper presentation)

Apr 2024 Comparing the Spatial Patterns of Rainfall and Atmospheric Moisture among Three Major Tropical Cyclones. **AAG Annual Meeting, Honolulu, HI.** (Poster)

Nov 2023 Comparing Spatial Structure of Total Rainfall for Three Major Tropical Cyclones by Using IMERG Data. **Southeast Division of the AAG (SEDAAG)**, **Norfolk, VA** (Poster)

Aug 2022 Mapping for Women's Evacuation Plans During Climate-Induced Disasters **State of the Map, Florence, Italy** (Lightning talk)

Jul 2021 Why Does Sexism within OpenStreetMap Matter? **State of the Map (online)**. (Panel and workshop session)

Nov 2021 Importance of Advancing Geographic and Geospatial Knowledge in Women. **Geography 2050, The American Geographical Society (online)**.

Nov 2019 Panel discussion on how women are advancing in geospatial technology. **State of the Map Asia, Dhaka, Bangladesh.**

Nov 2018 Evacuation Plan during Cyclone Warning to Reduce Vulnerability of Women. **State of the Map Asia, Bengaluru, India.** (Lightning talk)

TRAINING

AI Applications Using Environmental Satellite Remote Sensing Data June 2026
American Meteorological Society (AMS) Committee on Satellite Meteorology, Oceanography, and Climatology (SatMOC) virtual short course;completed a four-session series

Introduction to Environmental Satellite Remote Sensing

Creating AI-Ready Datasets

Forest for the Trees: Understanding Machine Learning in Land Cover Classification

From Random Forests to Foundation Models: AI for Atmospheric Applications

Research Training, YouthMappers Research Fellowship June 2018
The George Washington University and West Virginia University, USA

Customized technical training and discussions on research methods and data analysis.

GIS for Project Planning, Monitoring and Management 2017
NGH & CRI, Global Health and Climate Institute, Department of Environmental Science and Management, North South University

Project

Community Risk Assessment (CRA), Cox's Bazar, Bangladesh October 2019
Knowledge, Information and Learning for Local Adaptation (KILLA); funded by UNDP
Survey Team Leader (Field Work)

TECHNICAL SKILLS

Programming and Data Analysis: Python (xarray, pandas, matplotlib, seaborn; Anaconda), R Studio, Visual Studio, NetCDF processing

GIS and Mapping: ArcGIS Pro, QGIS, Open Data Kit (ODK), OpenStreetMap tools (OSMand, OSM Trackers), InaSAFE

Datasets: IMERG V07, MSWEP V2.8, ERA5 (Copernicus CDS), TC PRIMED, IBTrACS

Others

Articles

- Shining Beyond the Bias: Everywhere She Maps. YouthMappers Blog. https://www.youthmappers.org/post/shining-beyond-the-bias-everywhere-she-maps-youthmappers
- An Interview with #EverywhereSheMaps Regional Ambassadors on women's participation in the geospatial industry and communities that use geospatial technologies. YouthMappers Blog https://www.youthmappers.org/post/an-interview-with-everywhereshemaps-regional-ambassadors